Does deflated Sharpe calculation correctly adjust for N=179 multiple testing?

Yes, the **Deflated Sharpe Ratio** correctly adjusts for multiple testing when the true trial count is used, but its validity depends entirely on accurate trial enumeration and distributional assumptions.

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The Deflated Sharpe Ratio was specifically designed to correct for selection bias arising from multiple testing and non-normally distributed returns, making it conceptually appropriate for a 179-trial search (Bailey & De Prado, 2014; Prado, 2014). Not controlling for the number of trials in a discovery process leads to over-optimistic performance expectations, and the DSR addresses this by deflating observed Sharpe ratios according to the number of strategies tested (Bailey & De Prado, 2014).

How DSR Handles Multiple Testing

The DSR corrects for two distinct sources of performance inflation: selection bias under multiple testing and non-normal return distributions (Bailey & De Prado, 2014; Prado, 2014). A related framework by Harvey and Liu provides a haircut method that determines the appropriate discount for any reported Sharpe ratio given the number of strategies explored, yielding a profit hurdle that strategies must clear to be deemed significant (Harvey & Liu, 2015). A more recent extension identifies five recurring pitfalls in Sharpe ratio inference, including failing to correct for multiple testing and selection effects, and proposes a closed-form approximation to the sampling distribution under non-normal and serially correlated returns (De Prado et al., 2026).

When researchers explore multiple strategies, the observed Sharpe ratio is overstated because the analyst effectively conducted multiple attempts to obtain a favorable result (Koshiyama & Firoozye, 2019). For example, with 10 independent trials and a single-test p-value of 0.05, the multiple-testing-adjusted p-value rises to 0.401, implying a 40% probability of finding a strategy at least as profitable as the observed one by chance (Koshiyama & Firoozye, 2019). The resulting haircut Sharpe ratio is necessarily smaller than the original estimate because the adjusted p-value exceeds the single-test p-value (Koshiyama & Firoozye, 2019).

Critical Assumptions and Limitations

Assumption	Impact on N=179 Validity	Source
Trial independence	The standard haircut formula assumes independent tests; dependent strategies require FWER or FDR methods instead (Koshiyama & Firoozye, 2019)	Koshiyama & Firoozye (2019)
Accurate trial count	If the true count exceeds 179 (e.g., unrecorded exploratory runs), the deflation is insufficient (Bailey & De Prado, 2014)	Bailey & López de Prado (2014)
Non-normality correction	Returns must exhibit skewness and kurtosis for the DSR to outperform classical t-statistics (De Prado et al., 2026)	López de Prado et al. (2026)

Assumption	Impact on N=179 Validity	Source
FWER vs. FDR choice	Bonferroni and Holm corrections control family-wise error rate but lose power with many comparisons; FDR methods may be more appropriate for large N (Menyhárt et al., 2021; Koshiyama & Firoozye, 2019)	Menyhárt et al. (2021)

FIGURE 1 Key assumptions governing DSR validity at N=179

A fundamental concern is that when test statistics are dependent — as they inevitably are when strategies share underlying data or signal construction — the simple independence-based adjustment no longer applies, and the joint distribution of the n test statistics must be considered (Koshiyama & Firoozye, 2019). In that more realistic case, researchers recommend techniques controlling the family-wise error rate (Bonferroni, Holm) or the false discovery rate (BHY procedure) (Koshiyama & Firoozye, 2019). However, Holm and Hochberg corrections are beneficial only when the number of comparisons is relatively low and the effect rate is high, and are not appropriate for correcting thousands of comparisons (Menyhárt et al., 2021).

Alternative Multiple Testing Frameworks

Even without correction, standard analysis pipelines produce at least one statistically significant estimate in more than 90% of experimental trials when none of the treatments has any real effect (Liu & Shiraito, 2023). All major correction methods — Bonferroni, Benjamini-Hochberg, and adaptive shrinkage — are more accurate in recovering the truth than conventional analysis without correction (Liu & Shiraito, 2023). The Bonferroni correction guards against false positives but incurs a significant cost in false negatives (Liu & Shiraito, 2023), while the Benjamini-Hochberg procedure is least susceptible to false negatives but most lenient with false positives (Liu & Shiraito, 2023).

In financial applications specifically, the false discovery rate method has been used to evaluate thousands of technical trading rules while keeping the proportion of false discoveries at a pre-specified level (Zarrabi et al., 2017; Psaradellis et al., 2021). The FDR approach estimates the number of lucky rules in the right tail of the performance distribution at a given significance level, providing a data-snooping control that is directly applicable to large strategy universes (Psaradellis et al., 2021).

Verdict for N=179

The DSR framework is **conceptually correct** for adjusting a Sharpe ratio after 179 trials, provided that three conditions hold: the trial count of 179 is complete and not undercounted, the dependence structure among the 179 strategies is either negligible or explicitly modeled, and the return distribution's non-normality parameters are accurately estimated. If any of these conditions fail — particularly if unrecorded exploratory runs mean the true N exceeds 179, or if the 179 strategies are highly correlated — the deflation will be insufficient and surviving strategies may still be statistical artifacts rather than genuine discoveries (Bailey & De Prado, 2014; Koshiyama & Firoozye, 2019; De Prado et al., 2026).

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